

Paper Code: MATMJ42 — Notes by Aryan

TENSOR ANALYSIS

Complete Professional Lecture Notes & Solved Manual

Rigorous Mathematical Analysis, Step-by-Step Proofs & Exam Solved Problems

Covering Definitions, Transformation Laws, Tensor Algebra, Riemannian Metric, Christoffel Symbols & Covariant Differentiation

Course Highlights Included in This Document:

- **Paper Code:** MATMJ42 (Tensor Analysis / Mathematical Physics)
- **Author:** Notes by Aryan
- **Complete Theory & Concepts:** All 23 Video Lectures rigorously transcribed and organized.
- **Step-by-Step Proofs:** Transformation laws, quotient law, tensor algebra, non-tensor character of Christoffel symbols, Ricci's theorem.
- **Solved Exam Questions:** Riemannian metric applications, Christoffel symbol calculations, index raising/lowering, curl of covariant vector.
- **Standard Notations:** Einstein summation convention, contravariant & covariant indices, metric tensors g_{ij}, g^{ij} .

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1 Unit 1: Introduction to Tensors & Index Conventions (Lectures 1–2)

1.1 Physical Motivation and Definition of Tensors (Lecture 1)

In physics and mathematics, quantities are classified according to how their components change under a transformation of coordinate systems:

1. **Scalars (Rank 0 Tensors)**: Quantities specified completely by a single real number (e.g., mass, temperature, scalar potential). A scalar ϕ has the exact same value in all coordinate systems: $\phi'(x') = \phi(x)$.
2. **Vectors (Rank 1 Tensors)**: Quantities requiring n components in an n -dimensional space (e.g., displacement, velocity, force). Their components transform linearly under a change of basis.
3. **Tensors of Rank $r \geq 2$** : Quantities having n^r components in n -dimensional space (e.g., stress tensor, inertia tensor, metric tensor, electromagnetic field tensor) that transform according to specific multi-linear transformation rules.

Definition: Tensor

A **tensor** is a geometrical or physical entity described by a set of components in a given coordinate system that transform according to specific linear homogeneous transformation laws under a change of coordinate systems.

1.2 Coordinate Systems and Index Placement

Let $x^1, x^2, \dots, x^n$ denote the coordinates of a point in an n -dimensional space in a system S , and $x'^1, x'^2, \dots, x'^n$ denote the coordinates of the same point in a transformed system S' .

Important Key Note / Exam Point: Superscripts vs. Subscripts

- **Superscript (x^i)**: Indicates a **Contravariant** index (e.g., x^1, x^2, x^3 are coordinates, NOT powers of x).
- **Subscript (A_i)**: Indicates a **Covariant** index.

1.3 Einstein Summation Convention (Lecture 2)

Definition: Einstein Summation Convention

If an index occurs **twice** in a single term—once as a superscript and once as a subscript—it implies a summation over that index from 1 to n , without writing the summation symbol $\sum$.

$$A_i B^i = \sum_{i=1}^n A_i B^i = A_1 B^1 + A_2 B^2 + \dots + A_n B^n$$

Important Key Note / Exam Point: Dummy Index vs. Free Index

1. **Dummy Index (Summation Index)**: An index repeated once upper and once lower. Its symbol can be replaced freely by any other unused letter without changing the expression value:

$$A_i B^i = A_j B^j = A_k B^k$$

2. **Free Index**: An index that appears only once in a given term. It must appear in every term of an equation at the same level (upper or lower).

Important Key Note / Exam Point: Rules of Tensor Algebra

- An index can NEVER appear more than twice in a single term. Expressions like $A_i B^i C_i$ are **meaningless and invalid**.
- Free indices on both sides of an equation must match in position and name.

1.4 The Kronecker Delta**Definition: Kronecker Delta δ_j^i**

The **Kronecker Delta** is defined as:

$$\delta_j^i = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

It is a mixed tensor of rank 2 (invariant numerical tensor).

Theorem / Law: Substitution Property of Kronecker Delta

Multiplying any tensor component by δ_j^i replaces the index j with i :

$$\delta_j^i A^j = A^i, \quad \delta_j^i A_i = A_j, \quad \delta_j^i B^{jk} = B^{ik}$$

Proof / Solution: Substitution Property

Expand the sum over $j = 1, 2, \dots, n$:

$$\delta_j^i A^j = \delta_1^i A^1 + \delta_2^i A^2 + \dots + \delta_i^i A^i + \dots + \delta_n^i A^n$$

Since $\delta_j^i = 0$ for all $j \neq i$ and $\delta_i^i = 1$, only the term $j = i$ survives, yielding $1 \cdot A^i = A^i$.

Important Key Note / Exam Point: Contraction of Kronecker Delta

In an n -dimensional space:

$$\delta_i^i = \delta_1^1 + \delta_2^2 + \dots + \delta_n^n = \underbrace{1 + 1 + \dots + 1}_{n \text{ times}} = n$$

2 Unit 2: Contravariant, Covariant, and Mixed Tensors (Lectures 3–9)

2.1 Contravariant Tensors and Transformation Law (Lectures 3, 4, 6)

Consider differential coordinate displacements dx^i in system S . By the chain rule of partial differentiation, the differentials in S' are:

$$dx'^i = \frac{\partial x'^i}{\partial x^j} dx^j$$

Quantities transforming like dx^i are called contravariant vectors.

Definition: Contravariant Tensor of Rank 1 (Contravariant Vector)

A set of n components A^i in system S is said to be a **contravariant vector** if its components A'^i in system S' are related by:

$$A'^i = \frac{\partial x'^i}{\partial x^j} A^j$$

Definition: Contravariant Tensor of Rank r

A set of n^r components $A^{i_1 i_2 \dots i_r}$ is a **contravariant tensor of rank r** if it transforms as:

$$A'^{i_1 i_2 \dots i_r} = \frac{\partial x'^{i_1}}{\partial x^{j_1}} \frac{\partial x'^{i_2}}{\partial x^{j_2}} \dots \frac{\partial x'^{i_r}}{\partial x^{j_r}} A^{j_1 j_2 \dots j_r}$$

For a contravariant tensor of rank 2:

$$A'^{ij} = \frac{\partial x'^i}{\partial x^k} \frac{\partial x'^j}{\partial x^l} A^{kl}$$

2.2 Covariant Tensors and Transformation Law (Lectures 5, 7)

Consider the gradient of a scalar field $\phi(x)$, $\frac{\partial \phi}{\partial x^i}$. Under coordinate transformation:

$$\frac{\partial \phi}{\partial x'^i} = \frac{\partial \phi}{\partial x^j} \frac{\partial x^j}{\partial x'^i}$$

Quantities transforming like the gradient of a scalar are called covariant vectors.

Definition: Covariant Tensor of Rank 1 (Covariant Vector)

A set of n components A_i is a **covariant vector** if its components A'_i in system S' satisfy:

$$A'_i = \frac{\partial x^j}{\partial x'^i} A_j$$

Definition: Covariant Tensor of Rank s

A set of n^s components $A_{j_1 j_2 \dots j_s}$ is a **covariant tensor of rank s** if:

$$A'_{j_1 j_2 \dots j_s} = \frac{\partial x^{k_1}}{\partial x'^{j_1}} \frac{\partial x^{k_2}}{\partial x'^{j_2}} \dots \frac{\partial x^{k_s}}{\partial x'^{j_s}} A_{k_1 k_2 \dots k_s}$$

For a covariant tensor of rank 2:

$$A'_{ij} = \frac{\partial x^k}{\partial x'^i} \frac{\partial x^l}{\partial x'^j} A_{kl}$$

2.3 Mixed Tensors and General Transformation Law (Lectures 8, 9)

Definition: Mixed Tensor of Rank $(r + s)$

A set of n^{r+s} components $A_{j_1 j_2 \dots j_s}^{i_1 i_2 \dots i_r}$ having r contravariant (upper) indices and s covariant (lower) indices is a **mixed tensor of rank $(r + s)$** (or type (r, s)) if:

$$A_{j_1 \dots j_s}^{i_1 \dots i_r} = \left(\frac{\partial x_1^{i_1}}{\partial x^{k_1}} \cdots \frac{\partial x_r^{i_r}}{\partial x^{k_r}} \right) \left(\frac{\partial x^{l_1}}{\partial x_1^{j_1}} \cdots \frac{\partial x^{l_s}}{\partial x_s^{j_s}} \right) A_{l_1 \dots l_s}^{k_1 \dots k_r}$$

For a mixed tensor of rank 2 (type $(1,1)$):

$$A_j^i = \frac{\partial x^{i'}}{\partial x^k} \frac{\partial x^l}{\partial x^{j'}} A_l^k$$

2.4 Group (Transitive) Property of Transformation Laws

Theorem / Law: Transitive Property of Tensor Transformations

If components transform as a tensor from system $S \rightarrow S'$, and from $S' \rightarrow S''$, then the direct transformation from $S \rightarrow S''$ satisfies the exact tensor transformation law.

Proof / Solution: Transitive Property for Contravariant Vector

Let $A^{i'} = \frac{\partial x^{i'}}{\partial x^j} A^j$ (from $S \rightarrow S'$) and $A''^k = \frac{\partial x''^k}{\partial x^{i'}} A^{i'}$ (from $S' \rightarrow S''$). Substitute $A^{i'}$ into A''^k :

$$A''^k = \frac{\partial x''^k}{\partial x^{i'}} \left(\frac{\partial x^{i'}}{\partial x^j} A^j \right) = \left(\frac{\partial x''^k}{\partial x^{i'}} \frac{\partial x^{i'}}{\partial x^j} \right) A^j = \frac{\partial x''^k}{\partial x^j} A^j$$

This is precisely the tensor transformation law directly from $S \rightarrow S''$. Thus, tensor transformations form a group.

Theorem / Law: Kronecker Delta is an Invariant Mixed Tensor

Show that δ_j^i is a mixed tensor of rank 2 and has identical components in all coordinate systems.

Proof / Solution: Tensor Character of Kronecker Delta

Apply the mixed tensor transformation law to δ_l^k :

$$\delta_j^i = \frac{\partial x^{i'}}{\partial x^k} \frac{\partial x^l}{\partial x^{j'}} \delta_l^k = \frac{\partial x^{i'}}{\partial x^k} \frac{\partial x^k}{\partial x^{j'}} \quad (\text{using substitution property on } l \rightarrow k)$$

By the chain rule of partial derivatives:

$$\frac{\partial x^{i'}}{\partial x^k} \frac{\partial x^k}{\partial x^{j'}} = \frac{\partial x^{i'}}{\partial x^{j'}} = \delta_j^i$$

Hence $\delta_j^i = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$. Thus, δ_j^i is an invariant mixed tensor of rank 2.

3 Unit 3: Symmetric & Antisymmetric Tensors (Lecture 10)

3.1 Definitions and Properties

Definition: Symmetric Tensor

A contravariant tensor A^{ij} (or covariant tensor A_{ij}) is **symmetric** with respect to indices i and j if:

$$A^{ij} = A^{ji} \quad \text{or} \quad A_{ij} = A_{ji}$$

Definition: Antisymmetric (Skew-Symmetric) Tensor

A contravariant tensor A^{ij} (or covariant tensor A_{ij}) is **antisymmetric** if:

$$A^{ij} = -A^{ji} \quad \text{or} \quad A_{ij} = -A_{ji}$$

For an antisymmetric tensor, all diagonal components vanish identically:

$$A^{ii} = -A^{ii} \implies 2A^{ii} = 0 \implies A^{ii} = 0 \quad (\text{no sum over } i)$$

3.2 Invariance of Symmetry Properties

Theorem / Law: Invariance of Symmetry Under Transformation

If a tensor is symmetric (or antisymmetric) in one coordinate system, it remains symmetric (or antisymmetric) in all coordinate systems.

Proof / Solution: Invariance of Symmetry

Let $A^{ij} = A^{ji}$ in system S . In system S' :

$$A'^{ij} = \frac{\partial x'^i}{\partial x^k} \frac{\partial x'^j}{\partial x^l} A^{kl}$$

Interchange the free dummy indices k and l (and use $A^{kl} = A^{lk}$):

$$A'^{ij} = \frac{\partial x'^i}{\partial x^l} \frac{\partial x'^j}{\partial x^k} A^{lk} = \frac{\partial x'^j}{\partial x^k} \frac{\partial x'^i}{\partial x^l} A^{kl} = A'^{ji}$$

Thus $A'^{ij} = A'^{ji}$, proving symmetry is an intrinsic coordinate-independent tensor property.

3.3 Number of Independent Components

In an n -dimensional space, a rank 2 tensor has n^2 total components.

Important Key Note / Exam Point: Component Counts**1. Symmetric Tensor:**

$$N_{\text{sym}} = \text{Diagonal components} + \frac{1}{2} \text{Off-diagonal components} = n + \frac{n^2 - n}{2} = \frac{n(n+1)}{2}$$

$$\text{For } n = 3: N_{\text{sym}} = \frac{3 \times 4}{2} = 6.$$

2. Antisymmetric Tensor:

$$N_{\text{anti}} = 0 \text{ (diagonal)} + \frac{n^2 - n}{2} = \frac{n(n-1)}{2}$$

$$\text{For } n = 3: N_{\text{anti}} = \frac{3 \times 2}{2} = 3.$$

3.4 Fundamental Tensor Decomposition Theorem**Theorem / Law: Tensor Decomposition Theorem**

Every second-order tensor A_{ij} can be uniquely expressed as the sum of a symmetric tensor S_{ij} and an antisymmetric tensor T_{ij} .

Proof / Solution: Decomposition Theorem

Define:

$$S_{ij} = \frac{1}{2}(A_{ij} + A_{ji}), \quad T_{ij} = \frac{1}{2}(A_{ij} - A_{ji})$$

Check symmetry of S_{ij} : $S_{ji} = \frac{1}{2}(A_{ji} + A_{ij}) = S_{ij}$ (Symmetric).

Check antisymmetry of T_{ij} : $T_{ji} = \frac{1}{2}(A_{ji} - A_{ij}) = -\frac{1}{2}(A_{ij} - A_{ji}) = -T_{ij}$ (Antisymmetric).

Sum $S_{ij} + T_{ij}$:

$$S_{ij} + T_{ij} = \frac{1}{2}A_{ij} + \frac{1}{2}A_{ji} + \frac{1}{2}A_{ij} - \frac{1}{2}A_{ji} = A_{ij}$$

Hence $A_{ij} = S_{ij} + T_{ij}$ uniquely.

4 Unit 4: Tensor Algebra & The Quotient Law (Lectures 11–14)

4.1 Addition and Subtraction of Tensors (Lecture 11)

Theorem / Law: Addition/Subtraction Theorem

The sum or difference of two tensors of the same rank and same type is another tensor of the exact same rank and type.

Proof / Solution: Addition Theorem

Let A_j^i and B_j^i be two mixed tensors of type (1,1). Under coordinate transformation:

$$A_j^i = \frac{\partial x'^i}{\partial x^k} \frac{\partial x^l}{\partial x'^j} A_l^k, \quad B_j^i = \frac{\partial x'^i}{\partial x^k} \frac{\partial x^l}{\partial x'^j} B_l^k$$

Add the two equations:

$$C_j^i = A_j^i + B_j^i = \frac{\partial x'^i}{\partial x^k} \frac{\partial x^l}{\partial x'^j} (A_l^k + B_l^k) = \frac{\partial x'^i}{\partial x^k} \frac{\partial x^l}{\partial x'^j} C_l^k$$

where $C_l^k = A_l^k + B_l^k$. Thus C_j^i is a mixed tensor of type (1,1).

4.2 Outer Product (Direct Product)

Theorem / Law: Outer Product Theorem

If A_j^i is a tensor of type (1,1) and B^{kl} is a tensor of type (2,0), then their outer product $C_j^{ikl} = A_j^i B^{kl}$ is a tensor of type $(1+2, 1+0) = (3, 1)$ of rank 4.

Proof / Solution: Outer Product Theorem

Write transformation equations:

$$A_j^i = \frac{\partial x'^i}{\partial x^m} \frac{\partial x^n}{\partial x'^j} A_n^m, \quad B^{kl} = \frac{\partial x'^k}{\partial x^p} \frac{\partial x^l}{\partial x'^q} B^{pq}$$

Multiply the two equations:

$$C_j^{ikl} = A_j^i B^{kl} = \frac{\partial x'^i}{\partial x^m} \frac{\partial x'^k}{\partial x^p} \frac{\partial x'^l}{\partial x^q} \frac{\partial x^n}{\partial x'^j} (A_n^m B^{pq}) = \frac{\partial x'^i}{\partial x^m} \frac{\partial x'^k}{\partial x^p} \frac{\partial x'^l}{\partial x^q} \frac{\partial x^n}{\partial x'^j} C_n^{mpq}$$

Thus C_j^{ikl} satisfies the transformation law of a type (3,1) tensor of rank 4.

4.3 Contraction of Tensors (Lecture 13)

Definition: Contraction

Contraction is the process of equating one contravariant index and one covariant index of a mixed tensor, implying a summation over that index.

Theorem / Law: Contraction Theorem

If a contraction is performed on a mixed tensor of rank $(r + s)$, the resulting quantity is a tensor of rank $(r - 1 + s - 1) = r + s - 2$.

Proof / Solution: Contraction Theorem

Let A^i_{jk} be a mixed tensor of type (1,2):

$$A^i_{jk} = \frac{\partial x'^i}{\partial x^l} \frac{\partial x^m}{\partial x'^j} \frac{\partial x^n}{\partial x'^k} A^l_{mn}$$

Set $k = i$ (contracting index i and k):

$$A^i_{ji} = \frac{\partial x'^i}{\partial x^l} \frac{\partial x^m}{\partial x'^j} \frac{\partial x^n}{\partial x'^i} A^l_{mn} = \left(\frac{\partial x'^i}{\partial x^l} \frac{\partial x^n}{\partial x'^i} \right) \frac{\partial x^m}{\partial x'^j} A^l_{mn} = \delta^n_l \frac{\partial x^m}{\partial x'^j} A^l_{mn} = \frac{\partial x^m}{\partial x'^j} A^l_{ml}$$

Let $B_j = A^i_{ji}$ and $B_m = A^l_{ml}$. Then $B'_j = \frac{\partial x^m}{\partial x'^j} B_m$, which is the transformation law of a covariant vector (rank 1 tensor). The rank has been reduced from 3 to 1 ($3 - 2 = 1$).

4.4 Inner Product of Tensors (Lecture 12)**Definition: Inner Product**

The **Inner Product** of two tensors is obtained by taking their outer product followed by a contraction over an upper index of one and a lower index of the other.

For example, the inner product of A^i_j and B^{jk} is $C^{ik} = A^i_j B^{jk}$, which is a contravariant tensor of rank 2 (type (2,0)).

4.5 Quotient Law of Tensors (Lecture 14)**Theorem / Law: Quotient Law of Tensors**

A set of components $A(i, j, k)$ whose inner product with an arbitrary tensor B^j yields a known tensor $C^i_k = A(i, j, k) B^j$ is itself a tensor A^i_{jk} .

Proof / Solution: Proof of Quotient Law

Let $A(i, j, k)$ be a set of quantities such that for any arbitrary contravariant vector B^j , the contraction $C_k^i = A(i, j, k)B^j$ is a mixed tensor of type (1,1).

Since C_k^i is a tensor:

$$C_k^i = \frac{\partial x'^i}{\partial x^m} \frac{\partial x^n}{\partial x'^k} C_n^m \implies A'(i, j, k)B'^j = \frac{\partial x'^i}{\partial x^m} \frac{\partial x^n}{\partial x'^k} A(m, p, n)B^p$$

Since B^j is a contravariant vector, $B^p = \frac{\partial x^p}{\partial x'^j} B'^j$. Substitute this into the right-hand side:

$$A'(i, j, k)B'^j = \frac{\partial x'^i}{\partial x^m} \frac{\partial x^n}{\partial x'^k} A(m, p, n) \left(\frac{\partial x^p}{\partial x'^j} B'^j \right)$$

Rearrange and factor out B'^j :

$$\left[A'(i, j, k) - \frac{\partial x'^i}{\partial x^m} \frac{\partial x^p}{\partial x'^j} \frac{\partial x^n}{\partial x'^k} A(m, p, n) \right] B'^j = 0$$

Since B^j (and thus B'^j) is completely **arbitrary**, the bracketed term must vanish identically:

$$A'(i, j, k) = \frac{\partial x'^i}{\partial x^m} \frac{\partial x^p}{\partial x'^j} \frac{\partial x^n}{\partial x'^k} A(m, p, n)$$

This is precisely the transformation law of a mixed tensor A_{jk}^i of type (1,2). Hence, $A(i, j, k)$ is a tensor A_{jk}^i .

5 Unit 5: Riemannian Metric & Index Gymnastics (Lectures 15–18)

5.1 Metric Tensor and Riemannian Line Element (Lecture 16)

In Riemannian geometry, the distance ds between two infinitesimally close points x^i and $x^i + dx^i$ is given by the quadratic differential form:

Definition: Riemannian Metric / Line Element

The **Riemannian metric** or **line element** ds^2 is defined as:

$$ds^2 = g_{ij} dx^i dx^j$$

where $g_{ij} = g_{ji}$ is a symmetric covariant tensor of rank 2, called the **Fundamental Metric Tensor**.

Important Key Note / Exam Point: Properties of Metric Tensor g_{ij}

1. Symmetric: $g_{ij} = g_{ji}$.
2. Non-singular: $g = \det(g_{ij}) \neq 0$.
3. Real and positive-definite ($ds^2 > 0$ for non-zero displacements in Riemannian space).

5.2 Reciprocal (Conjugate) Metric Tensor g^{ij} (Lecture 15)

Definition: Reciprocal Metric Tensor

The **reciprocal metric tensor** g^{ij} is defined as the matrix inverse of g_{ij} , satisfying:

$$g^{ij} g_{jk} = \delta_k^i$$

The components are given by:

$$g^{ij} = \frac{\text{Cofactor of } g_{ji} \text{ in } g}{g}, \quad \text{where } g = \det(g_{ij})$$

Theorem / Law: Tensor Property of g^{ij}

Prove that g^{ij} is a contravariant symmetric tensor of rank 2.

Proof / Solution: Reciprocal Metric Tensor

Since g_{ij} is a covariant tensor, $g_{ij} dx^i dx^j = ds^2$ is an invariant scalar. By the Quotient Law, since $g_{ij} A^i B^j$ is a scalar for arbitrary vectors A^i, B^j , g_{ij} is a tensor. Next, consider $g^{ij} g_{jk} = \delta_k^i$. Since δ_k^i is a tensor and g_{jk} is a tensor, by the Quotient Law, g^{ij} is a contravariant tensor of rank 2. Symmetry $g^{ij} = g^{ji}$ follows directly from the symmetry of g_{ij} .

5.3 Associated Vectors: Raising and Lowering Indices (Lecture 17)

The fundamental metric tensors g_{ij} and g^{ij} are used to convert contravariant components into covariant components and vice-versa.

Important Key Note / Exam Point: Index Raising and Lowering Operations

1. Lowering an Index:

$$A_i = g_{ij} A^j$$

2. Raising an Index:

$$A^i = g^{ij} A_j$$

3. Tensors of Higher Rank:

$$A^i_j = g^{ik} A_{kj}, \quad A_{ij} = g_{ik} g_{jl} A^{kl}, \quad A^{ij} = g^{ik} g^{jl} A_{kl}$$

5.4 Magnitude of a Vector and Angle Between Vectors**Definition: Magnitude (Length) of a Vector**

The square of the magnitude $\|A\|$ of a vector A^i is defined as:

$$\|A\|^2 = g_{ij} A^i A^j = A_i A^i = g^{ij} A_i A_j$$

A vector is a **unit vector** if $\|A\| = 1 \implies g_{ij} A^i A^j = 1$.

Definition: Angle Between Two Vectors

The cosine of the angle θ between two vectors A^i and B^i is defined by:

$$\cos \theta = \frac{g_{ij} A^i B^j}{\sqrt{g_{kl} A^k A^l} \sqrt{g_{mn} B^m B^n}} = \frac{A_i B^i}{\|A\| \|B\|}$$

Two vectors are **orthogonal** if:

$$g_{ij} A^i B^j = A_i B^i = 0 \implies \cos \theta = 0 \implies \theta = \frac{\pi}{2}$$

5.5 Solved Examples on Riemannian Metric (Lecture 18)**Solved Question / Problem: Metric Tensor in Polar Coordinates**

Find the components of the metric tensor g_{ij} , determinant g , and reciprocal metric tensor g^{ij} in 2D plane polar coordinates (r, θ) .

Proof / Solution: Solution for Polar Coordinates

The Euclidean line element in Cartesian coordinates is $ds^2 = dx^2 + dy^2$. Transformation to polar coordinates: $x = r \cos \theta$, $y = r \sin \theta$. Calculated differentials:

$$dx = \cos \theta dr - r \sin \theta d\theta, \quad dy = \sin \theta dr + r \cos \theta d\theta$$

Square and add:

$$ds^2 = (\cos \theta dr - r \sin \theta d\theta)^2 + (\sin \theta dr + r \cos \theta d\theta)^2 = dr^2 + r^2 d\theta^2$$

Let $x^1 = r$, $x^2 = \theta$. Comparing with $ds^2 = g_{11}(dx^1)^2 + 2g_{12}dx^1dx^2 + g_{22}(dx^2)^2$:

$$[g_{ij}] = \begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}$$

Determinant $g = \det(g_{ij}) = 1 \cdot r^2 - 0 = r^2$.

Reciprocal metric tensor $[g^{ij}] = [g_{ij}]^{-1}$:

$$[g^{ij}] = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{r^2} \end{pmatrix} \implies g^{11} = 1, \quad g^{22} = \frac{1}{r^2}, \quad g^{12} = g^{21} = 0$$

Solved Question / Problem: Important Proof Identity involving g_{ij} and g^{jk}

Prove that $g_{ij} \frac{\partial g^{jk}}{\partial x^k} = -g^{jk} \frac{\partial g_{ij}}{\partial x^k}$.

Proof / Solution: Solution

Differentiate the fundamental identity $g_{ij}g^{jk} = \delta_i^k$ with respect to x^k :

$$\frac{\partial}{\partial x^k} (g_{ij}g^{jk}) = \frac{\partial}{\partial x^k} (\delta_i^k) = 0$$

Apply product rule of differentiation:

$$\frac{\partial g_{ij}}{\partial x^k} g^{jk} + g_{ij} \frac{\partial g^{jk}}{\partial x^k} = 0 \implies g_{ij} \frac{\partial g^{jk}}{\partial x^k} = -g^{jk} \frac{\partial g_{ij}}{\partial x^k}$$

6 Unit 6: Christoffel Symbols & Non-Tensor Character (Lectures 19–22)

6.1 Definitions of Christoffel Symbols (Lectures 19, 20)

Christoffel symbols are fundamental mathematical objects constructed from partial derivatives of the metric tensor components g_{ij} .

Definition: Christoffel Symbol of First Kind $[ij, k]$

The **Christoffel symbol of the first kind** is defined as:

$$[ij, k] = \Gamma_{ijk} = \frac{1}{2} \left(\frac{\partial g_{ik}}{\partial x^j} + \frac{\partial g_{jk}}{\partial x^i} - \frac{\partial g_{ij}}{\partial x^k} \right)$$

Definition: Christoffel Symbol of Second Kind Γ_{ij}^k

The **Christoffel symbol of the second kind** is defined as:

$$\Gamma_{ij}^k = \left\{ \begin{matrix} k \\ ij \end{matrix} \right\} = g^{kl} [ij, l]$$

Multiplying by g_{km} gives the inverse relation: $[ij, m] = g_{km} \Gamma_{ij}^k$.

6.2 Symmetry Properties and Fundamental Relations

Important Key Note / Exam Point: Symmetry Properties

Christoffel symbols are symmetric with respect to their paired indices i and j :

$$[ij, k] = [ji, k] \quad \text{and} \quad \Gamma_{ij}^k = \Gamma_{ji}^k$$

Theorem / Law: Partial Derivative of Metric Tensor

Express $\frac{\partial g_{ij}}{\partial x^k}$ in terms of Christoffel symbols of the first kind.

Proof / Solution: Proof

Write definitions:

$$[ik, j] = \frac{1}{2} \left(\frac{\partial g_{ij}}{\partial x^k} + \frac{\partial g_{kj}}{\partial x^i} - \frac{\partial g_{ik}}{\partial x^j} \right), \quad [jk, i] = \frac{1}{2} \left(\frac{\partial g_{ji}}{\partial x^k} + \frac{\partial g_{ki}}{\partial x^j} - \frac{\partial g_{jk}}{\partial x^i} \right)$$

Add the two expressions (noting $g_{ij} = g_{ji}$):

$$[ik, j] + [jk, i] = \frac{1}{2} \left(2 \frac{\partial g_{ij}}{\partial x^k} + \frac{\partial g_{kj}}{\partial x^i} - \frac{\partial g_{jk}}{\partial x^i} + \frac{\partial g_{ki}}{\partial x^j} - \frac{\partial g_{ik}}{\partial x^j} \right) = \frac{\partial g_{ij}}{\partial x^k}$$

Hence:

$$\frac{\partial g_{ij}}{\partial x^k} = [ik, j] + [jk, i]$$

Theorem / Law: Derivative of Determinant g

Prove that $\Gamma_{ik}^i = \frac{\partial}{\partial x^k} (\ln \sqrt{g}) = \frac{1}{2g} \frac{\partial g}{\partial x^k}$.

Proof / Solution: Proof

Using Jacobi's formula for the derivative of a determinant: $\frac{\partial g}{\partial x^k} = gg^{ij} \frac{\partial g_{ij}}{\partial x^k}$.

Substitute $\frac{\partial g_{ij}}{\partial x^k} = [ik, j] + [jk, i]$:

$$\frac{\partial g}{\partial x^k} = gg^{ij} ([ik, j] + [jk, i]) = g (g^{ij} [ik, j] + g^{ij} [jk, i]) = g (\Gamma_{ik}^i + \Gamma_{jk}^j) = 2g\Gamma_{ik}^i$$

Dividing by $2g$:

$$\Gamma_{ik}^i = \frac{1}{2g} \frac{\partial g}{\partial x^k} = \frac{\partial}{\partial x^k} (\ln \sqrt{g})$$

6.3 Proof that Christoffel Symbols are NOT Tensors (Lectures 21, 22)**Theorem / Law: Transformation Law of Christoffel Symbols**

Prove that Christoffel symbols do NOT transform as tensors under general coordinate transformations.

Proof / Solution: Transformation Law Derivation

Start with the metric tensor transformation law $g'_{ij} = \frac{\partial x^m}{\partial x'^i} \frac{\partial x^n}{\partial x'^j} g_{mn}$.

Differentiate with respect to x'^k :

$$\frac{\partial g'_{ij}}{\partial x'^k} = \frac{\partial}{\partial x'^k} \left(\frac{\partial x^m}{\partial x'^i} \frac{\partial x^n}{\partial x'^j} g_{mn} \right) = \frac{\partial^2 x^m}{\partial x'^k \partial x'^i} \frac{\partial x^n}{\partial x'^j} g_{mn} + \frac{\partial x^m}{\partial x'^i} \frac{\partial^2 x^n}{\partial x'^k \partial x'^j} g_{mn} + \frac{\partial x^m}{\partial x'^i} \frac{\partial x^n}{\partial x'^j} \frac{\partial x^p}{\partial x'^k} \frac{\partial g_{mn}}{\partial x^p}$$

Forming the linear combination for $[ij, k]'$ yields:

$$[ij, k]' = \frac{\partial x^m}{\partial x'^i} \frac{\partial x^n}{\partial x'^j} \frac{\partial x^p}{\partial x'^k} [mn, p] + g_{mn} \frac{\partial x^n}{\partial x'^k} \frac{\partial^2 x^m}{\partial x'^i \partial x'^j}$$

Multiply by $g'^{kl} = \frac{\partial x'^k}{\partial x^r} \frac{\partial x'^l}{\partial x^s} g^{rs}$ to obtain Γ_{ij}^l :

$$\Gamma_{ij}^l = \frac{\partial x'^k}{\partial x^m} \frac{\partial x^p}{\partial x'^i} \frac{\partial x^q}{\partial x'^j} \Gamma_{pq}^m + \underbrace{\frac{\partial x'^k}{\partial x^m} \frac{\partial^2 x^m}{\partial x'^i \partial x'^j}}_{\text{Inhomogeneous Non-Tensor Term}}$$

Because of the second derivative term $\frac{\partial^2 x^m}{\partial x'^i \partial x'^j}$, Christoffel symbols are **NOT components of a tensor**. They are connection coefficients.

6.4 Solved Problems on Christoffel Symbols**Solved Question / Problem: Christoffel Symbols in 2D Polar Coordinates**

Calculate all non-vanishing Christoffel symbols of the first kind $[ij, k]$ and second kind Γ_{ij}^k for 2D polar coordinates ($x^1 = r, x^2 = \theta$).

Proof / Solution: Solution for 2D Polar Coordinates

Given metric tensor: $g_{11} = 1, g_{22} = r^2, g_{12} = g_{21} = 0$.

Derivatives of metric components: $\frac{\partial g_{22}}{\partial r} = \frac{\partial(r^2)}{\partial r} = 2r$. All other derivatives are zero.

1. **Christoffel Symbols of First Kind**:

$$\bullet [22, 1] = \frac{1}{2} \left(\frac{\partial g_{21}}{\partial \theta} + \frac{\partial g_{21}}{\partial \theta} - \frac{\partial g_{22}}{\partial r} \right) = \frac{1}{2}(0 + 0 - 2r) = -r.$$

$$\bullet [12, 2] = [21, 2] = \frac{1}{2} \left(\frac{\partial g_{12}}{\partial \theta} + \frac{\partial g_{22}}{\partial r} - \frac{\partial g_{12}}{\partial \theta} \right) = \frac{1}{2}(2r) = r.$$

2. **Christoffel Symbols of Second Kind** ($\Gamma_{ij}^k = g^{kk}[ij, k]$):

$$\bullet \Gamma_{22}^1 = g^{11}[22, 1] = 1 \cdot (-r) = -r.$$

$$\bullet \Gamma_{12}^2 = \Gamma_{21}^2 = g^{22}[12, 2] = \frac{1}{r^2} \cdot r = \frac{1}{r}.$$

All other Christoffel symbols vanish identically!

Solved Question / Problem: Curl of a Covariant Vector Field

Prove that $\nabla_j A_i - \nabla_i A_j = \frac{\partial A_i}{\partial x^j} - \frac{\partial A_j}{\partial x^i}$.

Proof / Solution: Solution

Write definitions of covariant derivatives:

$$\nabla_j A_i = \frac{\partial A_i}{\partial x^j} - \Gamma_{ij}^k A_k, \quad \nabla_i A_j = \frac{\partial A_j}{\partial x^i} - \Gamma_{ji}^k A_k$$

Subtract the two equations:

$$\nabla_j A_i - \nabla_i A_j = \left(\frac{\partial A_i}{\partial x^j} - \Gamma_{ij}^k A_k \right) - \left(\frac{\partial A_j}{\partial x^i} - \Gamma_{ji}^k A_k \right)$$

Since Christoffel symbols are symmetric ($\Gamma_{ij}^k = \Gamma_{ji}^k$), the terms containing Γ cancel out:

$$\nabla_j A_i - \nabla_i A_j = \frac{\partial A_i}{\partial x^j} - \frac{\partial A_j}{\partial x^i}$$

This proves that the generalised curl of a covariant vector is independent of Christoffel symbols and depends only on ordinary partial derivatives!

7 Unit 7: Covariant Differentiation & Ricci's Theorem (Lecture 23)

7.1 Need for Covariant Differentiation

In Cartesian coordinates, the partial derivative $\frac{\partial A^i}{\partial x^j}$ of a vector A^i is a tensor. However, in general curvilinear/curved space coordinates, the partial derivative fails to be a tensor because basis vectors vary from point to point.

Under coordinate transformation:

$$\frac{\partial A'^i}{\partial x'^j} = \frac{\partial}{\partial x'^j} \left(\frac{\partial x'^i}{\partial x^m} A^m \right) = \frac{\partial x'^i}{\partial x^m} \frac{\partial x^n}{\partial x'^j} \frac{\partial A^m}{\partial x^n} + \frac{\partial^2 x'^i}{\partial x^n \partial x^m} \frac{\partial x^n}{\partial x'^j} A^m$$

The inhomogeneous second derivative term destroys the tensor property. To fix this, we introduce the **Covariant Derivative**.

7.2 Definitions of Covariant Derivatives

Definition: Covariant Derivative of Contravariant Vector

The **covariant derivative** of a contravariant vector A^i with respect to x^j , denoted $A^i_{;j}$ or $\nabla_j A^i$, is defined as:

$$A^i_{;j} = \nabla_j A^i = \frac{\partial A^i}{\partial x^j} + \Gamma^i_{jk} A^k$$

Definition: Covariant Derivative of Covariant Vector

The **covariant derivative** of a covariant vector A_i with respect to x^j , denoted $A_{i;j}$ or $\nabla_j A_i$, is defined as:

$$A_{i;j} = \nabla_j A_i = \frac{\partial A_i}{\partial x^j} - \Gamma^k_{ij} A_k$$

Definition: Covariant Derivative of Higher-Rank Tensors

1. **Mixed Tensor A^i_j :**

$$A^i_{j;k} = \frac{\partial A^i_j}{\partial x^k} + \Gamma^i_{kl} A^l_j - \Gamma^l_{jk} A^i_l$$

2. **Covariant Tensor A_{ij} :**

$$A_{ij;k} = \frac{\partial A_{ij}}{\partial x^k} - \Gamma^l_{ik} A_{lj} - \Gamma^l_{jk} A_{il}$$

3. **Contravariant Tensor A^{ij} :**

$$A^{ij}_{;k} = \frac{\partial A^{ij}}{\partial x^k} + \Gamma^i_{kl} A^{lj} + \Gamma^j_{kl} A^{il}$$

Each contravariant index adds a $+\Gamma$ term, and each covariant index adds a $-\Gamma$ term.

7.3 Ricci's Theorem (Fundamental Theorem of Riemannian Geometry)

Theorem / Law: Ricci's Theorem

The covariant derivatives of the fundamental metric tensors g_{ij} , g^{ij} , and δ^i_j vanish identically:

$$g_{ij,k} = 0, \quad g^{ij}_{;k} = 0, \quad \delta^i_{j,k} = 0$$

Proof / Solution: Proof of Ricci's Theorem for g_{ij}

Apply the formula for covariant derivative of a rank 2 covariant tensor g_{ij} :

$$g_{ij,k} = \frac{\partial g_{ij}}{\partial x^k} - \Gamma_{ik}^l g_{lj} - \Gamma_{jk}^l g_{il}$$

Recall that $\Gamma_{ik}^l g_{lj} = [ik, j]$ and $\Gamma_{jk}^l g_{il} = [jk, i]$. Substitute these:

$$g_{ij,k} = \frac{\partial g_{ij}}{\partial x^k} - [ik, j] - [jk, i]$$

From Unit 6 Theorem 6.2, we know $\frac{\partial g_{ij}}{\partial x^k} = [ik, j] + [jk, i]$. Substitute this derivative:

$$g_{ij,k} = ([ik, j] + [jk, i]) - [ik, j] - [jk, i] = 0$$

Hence $g_{ij,k} = 0$. By raising indices, $g^{ij}_{,k} = 0$ and $\delta^i_{j,k} = 0$. This completes the proof of Ricci's Theorem.

Important Key Note / Exam Point: Consequence of Ricci's Theorem

Since $g_{ij,k} = 0$, the metric tensor behaves like a **constant** under covariant differentiation. Thus, metric tensors g_{ij} and g^{ij} can be passed inside or outside covariant derivatives freely when raising/lowering indices:

$$\nabla_k(g_{ij}A^j) = g_{ij}\nabla_kA^j = \nabla_kA_i$$

8 Summary & Quick Revision Cheat Sheet

Table 1: Master Tensor Summary & Quick Formula Reference

Concept / Object	Symbol	Transformation / Definition Formula
Contravariant Vector	A^i	$A'^i = \frac{\partial x'^i}{\partial x^j} A^j$
Covariant Vector	A_i	$A'_i = \frac{\partial x^j}{\partial x'^i} A_j$
Contravariant Tensor Rank 2	A^{ij}	$A'^{ij} = \frac{\partial x'^i}{\partial x^k} \frac{\partial x'^j}{\partial x^l} A^{kl}$
Covariant Tensor Rank 2	A_{ij}	$A'_{ij} = \frac{\partial x^k}{\partial x'^i} \frac{\partial x^l}{\partial x'^j} A_{kl}$
Mixed Tensor Type (1,1)	A^i_j	$A'^i_j = \frac{\partial x'^i}{\partial x^k} \frac{\partial x^l}{\partial x'^j} A^k_l$
Kronecker Delta	δ^i_j	$\delta^i_j = 1$ if $i = j$, 0 if $i \neq j$; $\delta^i_j A^j = A^i$
Symmetric Tensor Components	N_{sym}	$\frac{n(n+1)}{2}$
Antisymmetric Tensor Components	N_{anti}	$\frac{n(n-1)}{2}$
Riemannian Metric Line Element	ds^2	$ds^2 = g_{ij} dx^i dx^j$
Reciprocal Metric Tensor	g^{ij}	$g^{ij} g_{jk} = \delta^i_k$, $g^{ij} = \frac{\text{cofactor of } g_{ji}}{g}$
Index Lowering / Raising	A_i, A^i	$A_i = g_{ij} A^j$, $A^i = g^{ij} A_j$
Vector Magnitude	$\ A\ $	$\ A\ ^2 = g_{ij} A^i A^j = A_i A^i$
Christoffel Symbol 1st Kind	$[ij, k]$	$\frac{1}{2} \left(\frac{\partial g_{ik}}{\partial x^j} + \frac{\partial g_{jk}}{\partial x^i} - \frac{\partial g_{ij}}{\partial x^k} \right)$
Christoffel Symbol 2nd Kind	Γ^k_{ij}	$g^{kl} [ij, l]$
Covariant Deriv. (Contravariant)	$A^i_{;j}$	$\frac{\partial A^i}{\partial x^j} + \Gamma^i_{jk} A^k$
Covariant Deriv. (Covariant)	$A_{i;j}$	$\frac{\partial A_i}{\partial x^j} - \Gamma^k_{ij} A_k$
Ricci's Theorem	$g_{ij,k}$	$g_{ij,k} = 0$, $g^{ij}_{;k} = 0$